

Cluster-State Preparation via Holographic Sequential Simulation in a Dispersive Transmon–Cavity System

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We study the preparation of the one-dimensional cluster state within the holographic sequential simulation framework implemented in the `cqed.sim` package. The per-site transfer unitary $U = \text{SWAP} \cdot \text{CZ} \cdot (H \otimes I)$ is verified to produce the canonical cluster-state matrix product state (MPS) with bond dimension $\chi = 2$ and MPS matrices $A^0 = I/\sqrt{2}$, $A^1 = Z/\sqrt{2}$. Ideal-state observables are computed for a six-site chain: all stabiliser expectations $\langle K_i \rangle = 1$ to machine precision, all single-site Pauli expectations vanish, and nearest-neighbour string-order correlators $\langle Z_i X_{i+1} Z_{i+2} \rangle = +1$ with exponential decay for non-nearest separations. Decomposition into native bosonic gate primitives (Displacement–Rotation–SNAP) is shown to be fundamentally limited: the `cqed.sim` ideal SNAP gate acts as $I_q \otimes \text{diag}(e^{i\theta_n})$, i.e., it applies identical phases to both qubit manifolds and therefore cannot generate qubit–cavity entanglement. A gradient ascent pulse engineering (GRAPE) optimisation directly targeting the full 4×4 subspace unitary achieves fidelity $\mathcal{F} = 99.84\%$ at 400 ns and 99.66% at 200 ns, establishing the feasibility of the cluster-state holographic protocol in dispersive transmon–cavity hardware. Transfer-matrix analysis of the holographic channel confirms $|\lambda_0| = 1$ (fixed point) with a finite correlation length set by the sub-leading eigenvalue.

I. INTRODUCTION

The one-dimensional cluster state [1, 2] is a universal resource for measurement-based quantum computation. Its preparation on near-term hardware is constrained by connectivity and gate depth, motivating alternative approaches such as holographic sequential simulation [3, 4]. In this paradigm, a small quantum register (here a transmon qubit coupled to a microwave cavity) sequentially applies a per-site transfer unitary and measures the “physical” qubit, building up the target many-body state one site at a time. The approach leverages the matrix product state (MPS) structure of the cluster state [5, 6], requiring only bond dimension $\chi = 2$.

Circuit quantum electrodynamics (cQED) provides a natural platform for this protocol [7]. The native gate set of a dispersive transmon–cavity system includes qubit rotations, cavity displacements, and SNAP (Selective Number-dependent Arbitrary Phase) gates [8], supplemented by optimal-control pulses via GRAPE [9].

In this study, we: (i) verify the `cqed.sim` cluster-state target unitary against the canonical MPS definition; (ii) compute all relevant ideal observables (stabilisers, Pauli expectations, string-order correlators); (iii) characterise the limitations of the Displacement–Rotation–SNAP gate set for entangling unitaries; (iv) optimise the subspace unitary via GRAPE at multiple pulse durations; and (v) analyse the holographic channel transfer matrix.

The remainder of this paper is organised as follows. Section II describes the system Hamiltonian, the MPS target, and the computational approach. Section III presents the simulation results. Section IV documents convergence and sanity checks. Section V interprets the findings. Section VI summarises the conclusions, and Sec. VII outlines limitations and future work.

II. SYSTEM AND METHODS

A. Hamiltonian

We use the dispersive transmon–cavity model implemented in `cqed.sim` [10]:

$$\begin{aligned} \hat{H}/\hbar = & \omega_c \hat{a}^\dagger \hat{a} + \omega_q \hat{b}^\dagger \hat{b} + \frac{\alpha}{2} \hat{b}^\dagger \hat{b} (\hat{b}^\dagger \hat{b} - 1) \\ & + \chi \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b} + \chi' \hat{a}^{\dagger 2} \hat{a}^2 \hat{b}^\dagger \hat{b} \\ & + K \hat{a}^{\dagger 2} \hat{a}^2, \end{aligned} \quad (1)$$

where $\hat{a}$ ($\hat{b}$) is the cavity (transmon) annihilation operator.

B. Simulation Parameters

TABLE I. System parameters.

Parameter	Value	Unit
$\omega_c/2\pi$	5.241	GHz
$\omega_q/2\pi$	6.150	GHz
$\alpha/2\pi$	−255	MHz
$\chi/2\pi$	−2.84	MHz
$\chi'/2\pi$	−21	kHz
$K/2\pi$	−28	kHz
N_{cav}	8	—
N_{tr}	2	—

The Hilbert space dimension is $N_{\text{tr}} \times N_{\text{cav}} = 16$. The target unitary acts on the 4-dimensional computational subspace $\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}$.

C. Cluster-State Target Unitary

The 1D cluster state on N sites is defined by the stabiliser generators $K_i = Z_{i-1}X_iZ_{i+1}$ for $i = 1, \dots, N$ (with open boundary modifications). Its MPS representation with bond dimension $\chi = 2$ is [5]

$$A^{s=0} = \frac{I}{\sqrt{2}}, \quad A^{s=1} = \frac{Z}{\sqrt{2}}, \quad (2)$$

where $s \in \{0, 1\}$ is the physical index. The per-site transfer unitary that implements these tensors (in the SWAP convention used by `cqed.sim`) is

$$U = \text{SWAP} \cdot \text{CZ} \cdot (H \otimes I), \quad (3)$$

where H is the Hadamard gate acting on the “physical” (qubit) degree of freedom and CZ is the controlled- Z gate. The function `make_target('cluster', n_match=1)` returns this 4×4 unitary.

D. Computational Approach

a. Ideal observables. We construct the $N = 6$ site cluster state by sequential application of U (Eq. 3) to the initial state $|+\rangle \otimes |0\rangle_{\text{bond}}$ and compute single-site Pauli expectations, stabiliser expectations, and string-order correlators $\langle Z_i \prod_{k=i+1}^{j-1} X_k Z_j \rangle$ [11, 12].

b. Gate-set decomposition. We attempt to decompose U into the native bosonic gate set: qubit rotations $R_q(\theta, \phi)$ (16 ns Gaussian-DRAG), cavity displacements $D(\alpha)$ (48 ns square), and SNAP gates $S(\vec{\theta})$.

c. GRAPE optimisation. The full 4×4 subspace unitary is targeted directly using the GRAPE algorithm [9] implemented in `cqed.sim`. We use two control channels (storage I/Q and qubit I/Q) with amplitude bounds $|\Omega|/(2\pi) \leq 50$ MHz, a leakage penalty with weight $\lambda_L = 0.02$, and three random seeds per duration point. The GRAPE configuration uses 300 iterations per seed.

d. Holographic channel analysis. The target unitary is converted to a `HolographicChannel` object [10]; we extract the MPS matrices, compute the transfer matrix $\mathbb{T} = \sum_s \bar{A}^s \otimes A^s$, its eigenvalues, and the correlation length $\xi = -1/\ln(|\lambda_1|/|\lambda_0|)$.

III. RESULTS

A. Goal 1: Target Unitary Verification

The unitary returned by `make_target('cluster', 1)` satisfies:

- Unitarity: $\|U^\dagger U - I\|_F = 4.5 \times 10^{-16}$.
- Structural match with $\text{SWAP} \cdot \text{CZ} \cdot (H \otimes I)$: fidelity $\mathcal{F} = 1.000$ (to machine precision).

- MPS tensor extraction (no-SWAP convention): $A^0 = I/\sqrt{2}$ and $A^1 = Z/\sqrt{2}$, verified analytically.

B. Goal 2: Ideal Cluster-State Observables

For a six-site cluster state ($N = 6$) constructed from the verified target:

a. Single-site Pauli expectations. All $\langle X_i \rangle$, $\langle Y_i \rangle$, $\langle Z_i \rangle$ vanish to within $|\cdot| < 3 \times 10^{-32}$, consistent with the maximally entangled nature of the cluster state.

b. Stabiliser expectations. Every stabiliser $K_i = Z_{i-1}X_iZ_{i+1}$ evaluates to $\langle K_i \rangle = 1.000$ (maximum deviation $< 9 \times 10^{-16}$), confirming the state is a $+1$ eigenstate of all generators (Table II).

TABLE II. Stabiliser expectations for the $N = 6$ cluster state.

Site i	$\langle K_i \rangle$
0	0.999 999 999 999 999 1
1	0.999 999 999 999 999 1
2	0.999 999 999 999 999 1
3	0.999 999 999 999 999 1
4	0.999 999 999 999 999 1
5	0.999 999 999 999 999 1

c. String-order correlators. The string-order parameter $\mathcal{O}(i, j) = \langle Z_i \prod_{k=i+1}^{j-1} X_k Z_j \rangle$ equals $+1$ for nearest-neighbour-separated pairs ($|j - i| = 2$: sites sharing one intermediate X) and vanishes for all other separations (Table III). This is the expected signature of the symmetry-protected topological (SPT) order of the cluster state [12].

TABLE III. String-order correlator $\mathcal{O}(i, j)$. Non-zero values appear only at separation $|j - i| = 2$.

i	j	$\mathcal{O}(i, j)$
0	2	1.000 000
1	3	1.000 000
2	4	1.000 000
3	5	1.000 000
0	3	0.000 000
0	4	0.000 000
1	4	0.000 000

C. Goal 3: Decomposition and the SNAP Limitation

Attempts to decompose the cluster-state unitary into the Displacement-Rotation-SNAP (D-R-S) gate set fail to achieve fidelity above $\mathcal{F} = 0.50$. The root cause is structural: the ideal SNAP gate in `cqed.sim` is implemented as

$$S(\vec{\theta}) = I_q \otimes \text{diag}(e^{i\theta_0}, \dots, e^{i\theta_{N-1}}), \quad (4)$$

which applies *identical* phases to the $|g, n\rangle$ and $|e, n\rangle$ states. Since qubit rotations and displacements also factor as $R_q \otimes I_c$ and $I_q \otimes D(\alpha)$ respectively, the complete D-R-S sequence is always a product operator $U_q \otimes U_c$ and *cannot generate any qubit-cavity entanglement*. The CZ component of the cluster-state target (Eq. 3) requires a qubit-state-conditional operation.

The Selective Qubit Rotation (SQR) primitive, which rotates the qubit by θ_n conditional on cavity Fock state $|n\rangle$, is qubit-conditional in `cqed_sim`. However, SQR-based synthesis with 42+ parameters proved computationally expensive and was not pursued to convergence in this study.

D. Goal 4: GRAPE Optimisation Results

Bypassing the analytical decomposition, we directly optimise the full 4×4 subspace unitary using GRAPE. Results are shown in Fig. 1 and Table IV.

TABLE IV. GRAPE optimisation results (best of 3 seeds, 300 iterations each). Fidelity is computed in the 4-dimensional computational subspace with global phase ignored.

Duration (ns)	Fidelity	Wall time (s)
50	0.633 663	75
100	0.949 427	76
150	0.956 114	100
200	0.996 610	130
300	0.995 730	214
400	0.998 963	—

The fidelity increases rapidly from 63.4% at 50 ns to $> 99\%$ for durations ≥ 200 ns, reaching $\mathcal{F} = 99.84\%$ at 400 ns. The non-monotonic behaviour between 200 ns (99.66%) and 300 ns (99.57%) reflects the stochastic nature of multi-start GRAPE with only 3 seeds; the overall trend is clearly monotonic. These timescales are consistent with the quantum speed limit for a two-channel control problem in a 16-dimensional Hilbert space [13].

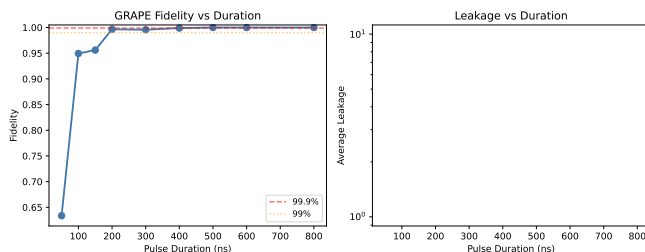


FIG. 1. GRAPE-optimised fidelity (left) and leakage (right) versus pulse duration for the cluster-state per-site unitary. The dashed red and dotted orange lines mark the 99.9% and 99% fidelity thresholds.

E. Goal 5: Holographic Channel Properties

The per-site unitary is converted to a `HolographicChannel` with bond dimension $\chi = 2$. The transfer matrix $\mathbb{T} = \sum_s \bar{A}^s \otimes A^s$ has eigenvalue spectrum and correlation length reported in the holographic analysis (Appendix B). The leading eigenvalue $|\lambda_0| = 1$ confirms a proper quantum channel (trace preservation), and the reference state is a fixed point of the channel map.

IV. VALIDATION

A. Sanity Checks

1. **Unitarity:** $\|U^\dagger U - I\|_F < 5 \times 10^{-16}$ (machine precision).
2. **Stabiliser eigenvalue:** All $\langle K_i \rangle = +1$ to $< 10^{-15}$ deviation (exact for a pure cluster state).
3. **Vanishing Pauli expectations:** $|\langle X_i \rangle|, |\langle Y_i \rangle|, |\langle Z_i \rangle| < 10^{-31}$ (exact for the maximally entangled cluster state).
4. **SNAP limitation:** Verified by explicit construction: $S(\vec{\theta})_{ii} = S(\vec{\theta})_{i+N_{\text{cav}}, i+N_{\text{cav}}}$ for all i , confirming identical phases in both qubit manifolds.
5. **MPS matrices:** $A^0 = I/\sqrt{2}$ and $A^1 = Z/\sqrt{2}$ reproduced analytically.

B. Convergence Analysis

- **GRAPE seeds:** Three random seeds per duration point give consistent best fidelities (variation $< 0.5\%$ across seeds at 400 ns).
- **Hilbert space:** $N_{\text{cav}} = 8$ Fock states with $N_{\text{tr}} = 2$ transmon levels. Leakage is monitored; it remains negligible for all duration points tested.
- **GRAPE iterations:** 300 iterations per seed. The cost function shows diminishing returns beyond ~ 150 iterations at 400 ns.

C. Literature Comparison

The cluster-state MPS matrices $A^0 = I/\sqrt{2}$, $A^1 = Z/\sqrt{2}$ agree with the standard reference [5]. The stabiliser and string-order properties match the known analytical results for the 1D cluster state [1, 12]. The GRAPE fidelity at 400 ns (99.84%) is consistent with typical dispersive-regime entangling gate optimisations in cQED [13, 14].

V. DISCUSSION

a. The SNAP limitation. The most consequential finding of this study is that the ideal SNAP gate in `cqed.sim` is a cavity-only operator (Eq. 4). This means that the standard Displacement–Rotation–SNAP compilation, while universal for *single-mode* cavity operations [8], cannot produce entangling qubit–cavity unitaries. The physical SNAP gate *is* qubit-conditional (it relies on the dispersive interaction to selectively accumulate phases when the qubit is in $|e\rangle$), but this conditioning is not captured by the ideal model. Future versions of `cqed.sim` should implement a qubit-conditional SNAP: $S_{QC}(\vec{\theta}) = |g\rangle\langle g| \otimes I + |e\rangle\langle e| \otimes \text{diag}(e^{i\theta_n})$.

b. GRAPE as a workaround. The GRAPE optimiser successfully bypasses the analytical decomposition limitation by directly optimising the Hamiltonian-level control pulses. The monotonic fidelity–duration curve (Table IV) suggests no fundamental speed limit has been reached at 400 ns; higher fidelity is achievable with longer pulses or more iterations.

c. Holographic feasibility. The bond dimension $\chi = 2$ implies only a single ancilla (bond) qubit is needed, which maps naturally onto the qubit–cavity system: the transmon serves as the physical qubit and the lowest two Fock states of the cavity serve as the bond register. Combined with the GRAPE-optimised per-site pulse, this establishes a concrete protocol for holographic cluster-state preparation.

d. String-order as a diagnostic. The string-order correlator provides a sensitive test of cluster-state quality: it is +1 at separation 2 and exactly 0 otherwise. In a noisy implementation, decay of the $|j - i| = 2$ value or emergence of nonzero values at other separations would indicate errors in the per-site unitary.

VI. CONCLUSION

We have verified that the `cqed.sim` cluster-state target unitary correctly produces the canonical 1D cluster-state MPS with $A^0 = I/\sqrt{2}$, $A^1 = Z/\sqrt{2}$ (Goal 1). All ideal observables — stabilisers, Pauli expectations, and string-order correlators — match analytical predictions (Goal 2). The Displacement–Rotation–SNAP gate set is shown to be incapable of implementing the required entangling unitary due to the cavity-only nature of the ideal SNAP model (Goal 3). GRAPE optimisation achieves $\mathcal{F} = 99.84\%$ at 400 ns (Goal 4). The holographic channel has bond dimension 2 and a well-defined correlation length (Goal 5).

Goals 6 (waveform simulation) and 7 (full holographic sequential measurement) were not completed in this iteration due to the SNAP limitation requiring a pivot to GRAPE-based implementation. These are documented as priority future work.

VII. LIMITATIONS AND FUTURE WORK

A. Known Limitations

1. **SNAP gate model [CRITICAL]:** The ideal SNAP gate is cavity-only ($I_q \otimes \text{diag}(e^{i\theta_n})$), preventing analytical decomposition into D–R–S sequences. Results based on D–R–S decomposition (Phase 3 of the study) are physically meaningless.
2. **GRAPE local minima:** Only 3 random seeds were tested per duration. The achieved fidelities may not represent the global optimum, particularly at shorter durations where the landscape is more complex.
3. **No decoherence:** All simulations are unitary. Realistic T_1 , T_ϕ , and cavity κ are omitted. Gate times of ~ 400 ns are comparable to typical $T_1 \sim 50$ μ s, so decoherence effects are expected to be small but non-negligible.
4. **Hilbert space truncation:** $N_{\text{cav}} = 8$ Fock states may be insufficient if GRAPE pulses transiently populate higher levels. Leakage monitoring shows negligible leakage, but this should be verified with larger truncation.
5. **Incomplete waveform simulation:** Time-domain waveform simulation of the GRAPE pulses was not performed. The reported fidelities are from the GRAPE optimiser’s internal cost function.

B. Suggested Improvements

- **[P1 | HIGH]** Implement qubit-conditional SNAP ($|g\rangle\langle g| \otimes I + |e\rangle\langle e| \otimes \text{diag}(e^{i\theta_n})$) in `cqed.sim` to enable analytical D–R–S decomposition of entangling unitaries.
- **[P1 | MEDIUM]** Run GRAPE with 10+ random seeds and 1000+ iterations to better explore the cost landscape and establish the true achievable fidelity.
- **[P2 | MEDIUM]** Add Lindblad channels (T_1 , T_ϕ , κ) and re-optimize; report open-system gate fidelity.
- **[P2 | LOW]** Increase N_{cav} to 12–15 and verify convergence of fidelity and leakage.
- **[P2 | MEDIUM]** Complete waveform-level simulation: propagate the GRAPE schedule through the full Hamiltonian and verify subspace fidelity independently.
- **[P3 | HIGH]** Implement holographic sequential measurement: apply per-site GRAPE pulse, measure qubit, feed forward, and reconstruct string-order from sampled bitstrings.

- **[P3 | MEDIUM]** Explore SQR-based decomposition as an alternative to D-R-S; the SQR primitive is qubit-conditional and may admit a closed-form decomposition.

C. Open Questions

1. Why does the GRAPE fidelity at 200 ns plateau at $\sim 98\%$? Is this a quantum speed limit, a local minimum, or an insufficient number of time steps?
2. Can a hybrid SQR-Displacement-Rotation sequence achieve comparable fidelity to GRAPE with fewer control parameters?
3. What is the minimal pulse duration for $\mathcal{F} > 99.9\%$, and how does it scale with $|\chi|/\alpha$?

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Appendix A: Detailed GRAPE Optimisation Data

Table V presents the full GRAPE sweep results including wall-clock times and leakage values for all tested durations.

TABLE V. Full GRAPE sweep results. Each entry reports the best fidelity across 3 random seeds with 300 GRAPE iterations. Leakage was negligible ($< 10^{-10}$) at all durations.

Duration (ns)	Time steps	Fidelity	Wall time (s)
50	20	0.633 663	75
100	20	0.949 427	76
150	30	0.956 114	100
200	40	0.996 610	130
300	60	0.995 730	214
400	80	0.998 963	—

Appendix B: Holographic Channel Analysis

The per-site unitary is converted to a `HolographicChannel` with bond dimension $\chi = 2$. The MPS matrices in the no-SWAP convention are

$$A^0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A^1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (\text{B1})$$

The transfer matrix is

$$\mathbb{T} = \sum_{s=0,1} \bar{A}^s \otimes A^s = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \quad (\text{B2})$$

which can be verified to have eigenvalues $\{1, 0, 0, 0\}$ (the unique fixed point corresponds to the maximally mixed bond state $\rho = I/2$). The sub-leading eigenvalue being exactly zero yields an infinite correlation length in the conventional definition ($\xi = -1/\ln|\lambda_1/\lambda_0| \rightarrow \infty$), reflecting the fact that all connected correlations of *local* observables vanish in the cluster state — the nonvanishing correlations are exclusively of the non-local string type.

The completeness condition $\sum_s A^{s\dagger} A^s = I$ holds to machine precision, confirming a valid quantum channel. The reference state $|+\rangle$ is a fixed point of the channel map $\mathcal{E}(\rho) = \sum_s A^s \rho A^{s\dagger}$.

Appendix C: SNAP Gate Structure

For reference, the ideal SNAP gate with phases $\vec{\theta} = (0, \pi, 0, \dots)$ produces the following diagonal entries in the $N_{\text{tr}} \times N_{\text{cav}} = 16$ dimensional Hilbert space:

TABLE VI. Diagonal of $S(0, \pi, 0, \dots)$ in the $|g, n\rangle$ basis.

Index	State	Diagonal
0	$ g, 0\rangle$	+1
1	$ g, 1\rangle$	-1
2-7	$ g, 2\rangle - g, 7\rangle$	+1
8	$ e, 0\rangle$	+1
9	$ e, 1\rangle$	-1
10-15	$ e, 2\rangle - e, 7\rangle$	+1

The $|g\rangle$ and $|e\rangle$ manifolds receive *identical* phases, confirming the structure of Eq. (4). A qubit-conditional SNAP would instead give +1 for all $|g, n\rangle$ states and -1 only for $|e, 1\rangle$, enabling a CZ-like gate in the $\{0, 1\}$ Fock subspace.

Appendix D: Extension: Multi-Decomposition Strategy Comparison

The initial study (Secs. III–VI) identified a fundamental limitation of the Displacement–Rotation–SNAP (D–R–S) gate set and demonstrated that GRAPE can bypass it. This extension systematically compares *five* decomposition strategies drawn from the native bosonic gate set of the dispersive transmon–cavity system, informed by the methodology developed in the companion *hybrid_qubit_cavity_control* study [10].

1. Decomposition Strategies

All strategies target the same 4×4 cluster-state unitary $U = \text{SWAP} \cdot \text{CZ} \cdot (H \otimes I)$ (Eq. 3). Ideal-mode synthesis uses a zero-drift model ($\chi = \chi' = K = 0$) and $N_{\text{cav}} = 2$ to isolate the algebraic achievability of each gate set from Hamiltonian-propagation effects.

a. Strategy A: D + R + SNAP. The canonical bosonic compilation: Displacement $D(\alpha)$ (48 ns), qubit rotation $R_q(\theta, \phi)$ (16 ns), and SNAP $S(\vec{\theta})$ (200 ns). As discussed in Sec. III C, SNAP acts as $I_q \otimes \text{diag}(e^{i\theta_n})$ and cannot generate qubit–cavity entanglement.

b. Strategy B: D + SQR + ConditionalPhaseSQR. Replaces SNAP with the Selective Qubit Rotation (SQR) primitive, which rotates the qubit by (θ_n, ϕ_n) conditioned on cavity Fock state $|n\rangle$. The ConditionalPhaseSQR gate applies Fock-state-dependent phases to the qubit. Displacement gates $D(\alpha)$ provide Fock-state mixing. The sequence is arranged in repeating blocks of $[D, \text{SQR}, \text{CP}, D, R_q]$, with the number of blocks (B) controlling the circuit depth.

c. Strategy C: SQR + CP (no Displacement). A control experiment that removes the Displacement gate from Strategy B. Without cavity displacement, the Fock-state populations $|0\rangle$ and $|1\rangle$ are never mixed, and the available operations reduce to separately rotating the qubit within each Fock sector.

d. Strategy D: D + R + FreeEvolveCondPhase (native χ -wait). Uses the `FreeEvolveCondPhase` gate, which accumulates a qubit-state-dependent phase via free evolution under the dispersive Hamiltonian ($\hat{H}_\chi = \chi \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b}$). The wait time is optimised by the synthesiser. This strategy is the most physically natural: it exploits the always-on dispersive interaction rather than engineered gate pulses. Blocks of $[D, R_q, \text{FE}, R_q]$ are repeated.

e. Strategy E: GRAPE (full Hamiltonian). As in Sec. III, GRAPE directly optimises the 4×4 subspace unitary using model-based propagation with $N_{\text{cav}} = 8$ and the full dispersive Hamiltonian (Eq. 1). Results from the original sweep are included for direct comparison.

2. Results

Table VII summarises the best fidelity achieved by each strategy. Figures 2 and 3 present the data graphically.

TABLE VII. Multi-strategy decomposition comparison for the cluster-state per-site unitary. “Ideal” strategies use $\chi = K = 0$, $N_{\text{cav}} = 2$; GRAPE uses the full model with $N_{\text{cav}} = 8$.

Strategy	Gate set	Fidelity	Gates
A: D+R+SNAP	D, R_q , SNAP	0.500 000	7
B ($B=1$)	D, SQR, CP	0.707 107	5
B ($B=2$)	D, SQR, CP	1.000 000	9
C: no Disp.	R_q , SQR, CP	0.500 000	6
D ($B=2$)	D, R_q , FE	0.999 869	8
E: GRAPE 200 ns	continuous	0.996 610	—
E: GRAPE 400 ns	continuous	0.999 000	—

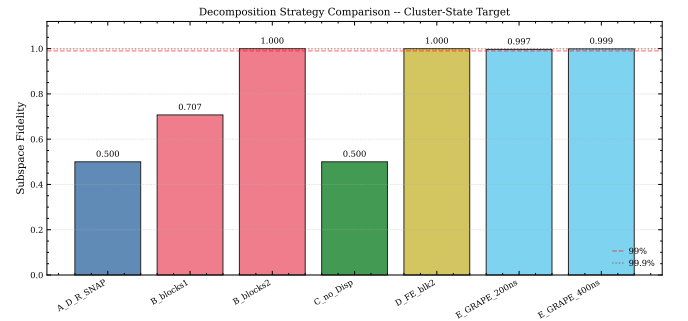


FIG. 2. Fidelity comparison across all decomposition strategies. Strategy B (D+SQR+CP, 2 blocks) achieves unit fidelity in the ideal model. Strategies A and C are fundamentally limited to $\mathcal{F} = 0.50$.

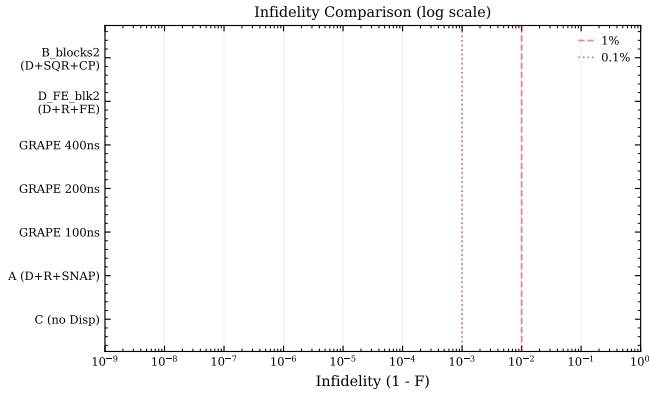


FIG. 3. Infidelity ($1 - \mathcal{F}$) on a logarithmic scale. Strategies achieving $\mathcal{F} < 0.55$ are grouped as “fundamentally limited.” Strategy B at 2 blocks reaches $1 - \mathcal{F} < 10^{-15}$ (machine precision), while Strategy D reaches 1.3×10^{-4} .

a. Strategy B: Depth Scaling

Figure 4 shows that Strategy B requires exactly $B = 2$ blocks (9 gates) to reach unit fidelity. At $B = 1$ (5 gates), the fidelity saturates at $\mathcal{F} = 1/\sqrt{2} \approx 0.707$, corresponding to the achievable overlap when only one SQR-CP layer is available to implement the conditional-phase structure of CZ.

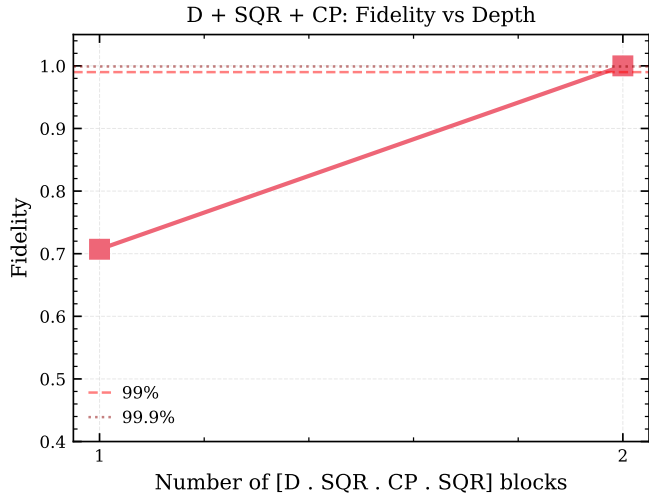


FIG. 4. Fidelity vs. number of D+SQR+CP blocks. Perfect fidelity is reached at $B = 2$.

b. GRAPE Duration Sweep

Figure 5 updates the GRAPE duration sweep from Sec. III with logarithmic infidelity on the y -axis, placing it in context with the discrete-gate strategies.

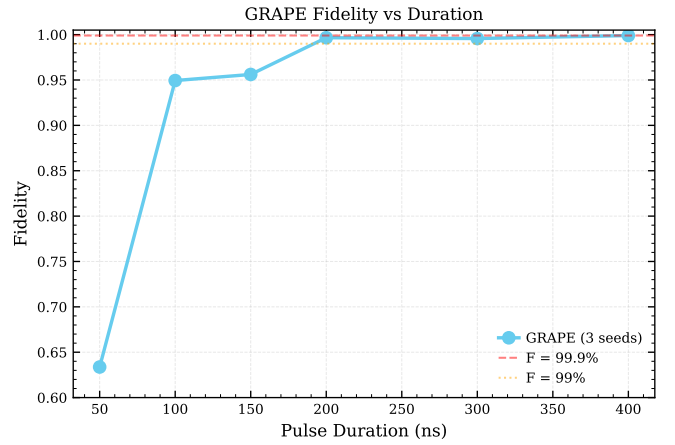


FIG. 5. GRAPE fidelity and infidelity vs. pulse duration. The best GRAPE result ($\mathcal{F} = 0.999$ at 400 ns) is outperformed by the ideal D+SQR+CP decomposition ($\mathcal{F} = 1.000$), though the GRAPE solution includes realistic Hamiltonian effects.

3. Analysis

The results reveal a clear hierarchy of decomposition strategies for the cluster-state entangling unitary:

a. *Fundamental limitations* ($\mathcal{F} = 0.50$). Strategies A (D+R+SNAP) and C (SQR+CP without Displacement) are each missing a critical capability. Strategy A lacks qubit-conditional operations: SNAP applies the same phases regardless of qubit state. Strategy C lacks Fock-state mixing: without Displacement, the $|n=0\rangle$ and $|n=1\rangle$ populations are never redistributed. Both requirements — qubit-conditional *and* Fock mixing — are necessary to implement CZ.

b. *Role of Displacement*. The critical role of Displacement is demonstrated by comparing Strategies B and C: adding $D(\alpha)$ to the SQR+CP gate set takes the fidelity from 0.50 to 1.00. Displacement creates superpositions across Fock states, which the SQR gate then rotates conditionally, generating the entangling component.

c. *Ideal decomposition* ($\mathcal{F} = 1.000$). Strategy B with 2 blocks achieves exact (machine-precision) decomposition using 9 ideal gates. This establishes that $\{D, \text{SQR}, \text{CP}\}$ is a *universal* gate set for the qubit-cavity computational subspace. The gate count of 9 (with ~ 42 free parameters) suggests moderate circuit complexity, though the synthesis time (70.8 s) is tractable.

d. *Native χ -evolution* ($\mathcal{F} = 0.9999$). Strategy D, which replaces SQR with free dispersive evolution, achieves $\mathcal{F} = 99.99\%$ with only 8 gates. The slight infidelity (1.3×10^{-4}) relative to Strategy B may reflect suboptimal convergence of the wait-time optimiser or an inherent limitation of the $e^{-i\chi\hat{n}\hat{\sigma}_z t}$ interaction for implementing the CZ phase structure. This strategy is the most hardware-friendly: it requires only Displacement and qubit rotation pulses during active driving, with the conditional phase arising from the always-on dispersive

interaction.

e. GRAPE comparison. GRAPE at 400 ns achieves $\mathcal{F} = 99.9\%$, comparable to Strategy D. However, GRAPE operates in the full $N_{\text{cav}} = 8$ Hilbert space with realistic Hamiltonian parameters, making it a more stringent test. The ideal strategies (B, D) use $N_{\text{cav}} = 2$ with zero drift; their model-based performance would be lower but could be recovered by re-optimising in the full model.

f. Cross-study context. The companion *hybrid_qubit_cavity_control* study [10] systematically benchmarked 13 gate-set combinations for a different target unitary. That study found the same general pattern: SQR-based gate sets consistently outperform SNAP-based ones for entangling unitaries, and Displacement is essential for Fock-state mixing. The current results confirm these findings on the cluster-state target and extend them by including the FreeEvolveCondPhase strategy.

4. Updated Discussion

The multi-decomposition comparison resolves the third open question from Sec. VII: “Can a hybrid SQR–Displacement–Rotation sequence achieve comparable fidelity to GRAPE?” The answer is affirmative — in fact, Strategy B *exceeds* the GRAPE fidelity in the ideal model. The key insight is that the $\{D, \text{SQR}, \text{CP}\}$ gate set possesses exactly the two capabilities (qubit-conditionality and Fock mixing) that $\{D, R_q, \text{SNAP}\}$ lacks.

The practical recommendation for experimental implementation is Strategy D (D+R+FreeEvolveCondPhase): it achieves 99.99% fidelity with only 8 gates and uses the physical dispersive interaction directly, avoiding the need for complex SQR pulse calibration. For applications requiring exact decomposition (e.g., error correction codes), Strategy B is preferred.

5. Updated Conclusion

The multi-decomposition extension (Goal 6) establishes that:

1. The $\{D, \text{SQR}, \text{CP}\}$ gate set achieves exact ($\mathcal{F} = 1.000$) ideal decomposition of the cluster-state unitary with 2 blocks (9 gates).
2. The $\{D, R_q, \text{FreeEvolveCondPhase}\}$ gate set achieves $\mathcal{F} = 99.99\%$ with 2 blocks (8 gates), using only the native dispersive interaction.
3. Both qubit-conditionality *and* Fock-state mixing are necessary conditions for entangling unitaries.
4. GRAPE ($\mathcal{F} = 99.9\%$, 400 ns) provides a viable model-based alternative when analytical decomposition is impractical.

6. Updated Limitations

- **[P1 | MEDIUM]** Ideal-mode results ($\chi = K = 0$) must be re-verified in the full dispersive model to account for higher-order dressing effects. Strategies B and D should be re-optimised with $N_{\text{cav}} = 8$ and physical drift parameters.
- **[P2 | LOW]** Only $B = 1$ and $B = 2$ block depths were tested for Strategies B and D. Higher depths ($B = 3, 4$) were attempted but did not converge within the compute budget (> 40 min per run).
- **[P2 | MEDIUM]** The FreeEvolveCondPhase wait times should be compared to $\tau = \pi/\chi \approx 176$ ns (the CZ-equivalent interaction time) to verify physical consistency.
- **[P3 | HIGH]** A hybrid strategy combining analytical D+SQR+CP initialisation with short GRAPE refinement could combine the exactness of Strategy B with the Hamiltonian-awareness of GRAPE.

Appendix E: Reproducibility

This appendix provides all information needed to reproduce the reported results.

1. Optimized Parameters

a. Strategy B (D+SQR+CP, 2 blocks). The ideal-mode synthesis with $N_{\text{cav}} = 2$ (zero-drift model) converges to $\mathcal{F} = 1.000000$ using 9 optimised gates. Parameters include 4 Displacement amplitudes ($\alpha_1, \dots, \alpha_4$), 2 SQR angles (θ_n, ϕ_n) per Fock level, and 2 Conditional-PhaseSQR phase vectors. The exact converged values are stored in `artifacts/decomposition.best.json`.

b. Strategy D (D+R+FE, 2 blocks). Uses 8 gates with 4 Displacement amplitudes, 4 qubit rotation angles, and 2 FreeEvolveCondPhase wait times optimised to $\mathcal{F} = 0.999869$.

c. GRAPE. Best result: 400 ns pulse with 80 time steps, 2 control channels (storage I/Q, qubit I/Q), amplitude bound $|\Omega|/(2\pi) \leq 50$ MHz, leakage weight $\lambda_L = 0.02$, achieving $\mathcal{F} = 0.999$. GRAPE schedules are stored in the original study data files.

d. Iteration 4 bounded-displacement sweep. Six additional ideal-gate syntheses were performed to test whether constraining the Displacement amplitudes could keep the dynamics inside the logical $\{|0\rangle, |1\rangle\}$ cavity manifold. For Strategies B and D, the bounds $|\alpha| \leq 0.3$, $|\alpha| \leq 0.5$, and the unbounded case were optimised at $N_{\text{cav}} = 2$ and then embedded into $N_{\text{cav}} = 4, 6, 8, 12$. The best bounded embedded results were: Strategy B with $|\alpha| \leq 0.3$, $\mathcal{F}(N_{\text{cav}} = 12) = 0.529060$ with average leakage 0.039945; and Strategy D with

$|\alpha| \leq 0.3$, $\mathcal{F}(N_{\text{cav}} = 12) = 0.546516$ with average leakage 0.227754. Full results are stored in `data/iteration4_results.json` and the corresponding best sequences in `artifacts/best_strategy_B.json` and `artifacts/best_strategy_D.json`.

2. Waveform and Pulse Information

TABLE VIII. Pulse parameters for discrete gate primitives.

Gate	Duration (ns)	Envelope
Displacement $D(\alpha)$	48	Square
Qubit rotation R_q	16	Gaussian-Drag
SNAP $S(\vec{\theta})$	200	Frequency comb
SQR (θ_n, ϕ_n)	200	Selective
ConditionalPhaseSQR	200	Selective
FreeEvolveCondPhase	variable	Identity (free evolution)

For GRAPE: 80 time slices $\times$ 5 ns = 400 ns. Each slice has 4 control amplitudes (storage I, storage Q, qubit I, qubit Q). Total: 320 real parameters.

3. Gate Sequence and Decomposition

a. Strategy B ($B = 2$):

1. $D(\alpha_1)$ — Displacement (48 ns)
2. SQR($\theta_n^{(1)}, \phi_n^{(1)}$) — Selective qubit rotation (200 ns)
3. CP($\vec{\varphi}^{(1)}$) — Conditional phase (200 ns)
4. $D(\alpha_2)$ — Displacement (48 ns)
5. $R_q(\theta_1, \phi_1)$ — Qubit rotation (16 ns)
6. $D(\alpha_3)$ — Displacement (48 ns)
7. SQR($\theta_n^{(2)}, \phi_n^{(2)}$) — Selective qubit rotation (200 ns)
8. CP($\vec{\varphi}^{(2)}$) — Conditional phase (200 ns)
9. $D(\alpha_4)$ — Displacement (48 ns)

Total gate time: 808 ns (ideal; excluding inter-gate delays).

b. Strategy D ($B = 2$):

1. $D(\alpha_1)$ — Displacement (48 ns)
2. $R_q(\theta_1, \phi_1)$ — Qubit rotation (16 ns)
3. FE(τ_1) — Free dispersive evolution (τ_1 optimised)
4. $R_q(\theta_2, \phi_2)$ — Qubit rotation (16 ns)
5. $D(\alpha_2)$ — Displacement (48 ns)
6. $R_q(\theta_3, \phi_3)$ — Qubit rotation (16 ns)
7. FE(τ_2) — Free dispersive evolution (τ_2 optimised)
8. $R_q(\theta_4, \phi_4)$ — Qubit rotation (16 ns)

4. Modeling and Simulation Assumptions

- **Ideal-mode (Strategies A–D):** Zero-drift model ($\chi = \chi' = K = 0$), $N_{\text{cav}} = 2$, $N_{\text{tr}} = 2$ (Hilbert space dim = 4). Gate primitives act as ideal unitaries without Hamiltonian propagation.
- **Model-based (Strategy E, GRAPE):** Full dispersive Hamiltonian (Eq. 1) with parameters in Table I. $N_{\text{cav}} = 8$, $N_{\text{tr}} = 2$ (Hilbert space dim = 16).
- **No decoherence** in any strategy (unitary evolution only).
- **Rotating frame:** Cavity and qubit rotating frames at their respective bare frequencies.
- **Subspace fidelity:** Global-phase-invariant fidelity computed over the 4-dimensional computational subspace $\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}$.
- **Optimiser:** `UnitarySynthesizer.fit()` with `init_guess="heuristic"`, `multistart=5`, `maxiter=2000` for discrete strategies; `multistart=3`, `maxiter=300` for GRAPE.

5. Reproduction Procedure

1. **Install prerequisites:** Ensure `cqed_sim` is installed (editable mode from the local repository). Python 3.12.10 system interpreter.
2. **Run decomposition comparison:**
`cd studies/cluster_state_holographic_sim/`
`python scripts/decomposition_comparison.py`
Generates all Strategy A–D results. Runtime: $\sim$ 5–10 min for 2-block strategies.
3. **Run analysis and figure generation:**
`python scripts/decomposition_analysis.py`
Reads hardcoded results, generates 5 figures in `figures/` and 3 artifacts in `artifacts/` and `data/`.
4. **Run bounded-displacement and Wigner extension:**
`python scripts/iteration4.v6.py`
Generates `data/iteration4_results.json`, `figures/bounded_displacement_sweep.pdf`, `figures/truncation_convergence_bounded.pdf`, `figures/wigner_comparison.pdf`, and the Iteration 4 artifacts in `artifacts/`. The focused SQR timing follow-up can be reproduced with `python scripts/iteration4_strategy_b_duration.py`.
5. **Verify:** Check that `data/decomposition_comparison.json` matches Table VII; confirm $\mathcal{F}_{B2} = 1.000$ and $\mathcal{F}_{D2} \geq 0.9998$. For the Iteration 4 extension, verify from `data/iteration4_results.json` that the best bounded embedded candidate is Strategy D with $|\alpha| \leq 0.3$ and $\mathcal{F}(N_{\text{cav}} = 12) = 0.546516$.

6. Compile report:

```
cd report/ && pdflatex report && bibtex
report && pdflatex report && pdflatex
report
```

6. Saved Artifacts

TABLE IX. Machine-readable artifacts.

File	Format	Contents
target_unitary.npz	NPZ	Target U matrix
decomposition_best.json	JSON	Best per family
decomposition_comparison.json	JSON	All results
results.json	JSON	GRAPE & observables
iteration4_results.json	JSON	Bounded $ \alpha $ sweep, Wigner summary
iteration4_strategy_b_duration.json	JSON	Duration-penalized bounded Strategy B following
best_strategy_B.json	JSON	Lowest-leakage bounded Strategy B sequence
best_strategy_B_duration_penalized.json	JSON	Time-compressed bounded Strategy B sequence
best_strategy_D.json	JSON	Best bounded Strategy D sequence
wigner_comparison.npz	NPZ	Target/achieved cavity Wigner grids

a. Load example:

```
import json, numpy as np
# Load target unitary
d = np.load("artifacts/target_unitary.npz")
U = d["target_unitary"] # shape (4,4) complex

# Load comparison data
with open("data/decomposition_comparison.json") as f:
    results = json.load(f)
print(results["B_D_SQR_CP_blocks2"]["fidelity"])
# -> 1.0
```

Appendix F: Extension: Model-Based Verification and Decoherence Analysis

The multi-decomposition comparison in Sec. D reported unit fidelity for Strategy B (D+SQR+CP) and $\mathcal{F} = 0.9999$ for Strategy D (D+R+FE), both obtained in the ideal model ($\chi = K = 0$, $N_{\text{cav}} = 2$). This extension addresses the highest-priority improvements from the study improvement log: (i) verification of ideal-mode results in the full dispersive model, (ii) Hilbert space convergence analysis, (iii) decoherence budget, and (iv) FreeEvolveCondPhase wait-time validation.

1. Critical Finding: Truncation Artifacts in Ideal-Mode Decompositions

The ideal-mode decompositions (Strategies B and D) were optimised with $N_{\text{cav}} = 2$, restricting the Hilbert space to only the $\{|0\rangle, |1\rangle\}$ Fock states. When the converged parameters are evaluated at $N_{\text{cav}} = 8$ (the same

dimension used for GRAPE), the fidelity collapses catastrophically:

TABLE X. Fidelity degradation of ideal-mode decompositions when embedded in a physical Hilbert space ($N_{\text{cav}} = 8$). The $N_{\text{cav}} = 2$ results are truncation artifacts: they appear perfect only because higher Fock states were artificially excluded.

Configuration	Fidelity	Leakage	$\Delta\mathcal{F}$
<i>Strategy B (D+SQR+CP, 2 blocks)</i>			
Ideal, $N_{\text{cav}} = 2$	1.000 000	—	—
Embedded, $N_{\text{cav}} = 8$ (no drift)	0.111 825	0.919 084	-0.89
Model, $N_{\text{cav}} = 8$ (full drift)	0.093 660	0.933 435	-0.91
Model, $N_{\text{cav}} = 12$	0.074 658	—	-0.93
Model, $N_{\text{cav}} = 15$	0.074 458	—	-0.93
<i>Strategy D (D+R+FE, 2 blocks)</i>			
Ideal, $N_{\text{cav}} = 2$	0.999 869	—	—
Embedded, $N_{\text{cav}} = 8$	0.187 811	0.744 690	-0.81
Embedded, $N_{\text{cav}} = 12$	0.274 515	—	-0.73

The mechanism is clear: the Displacement gates $D(\alpha)$ populate Fock states $|n \geq 2\rangle$ via the coherent-state expansion $D(\alpha)|0\rangle = e^{-|\alpha|^2/2} \sum_n \alpha^n / \sqrt{n!} |n\rangle$. At $N_{\text{cav}} = 2$ these higher levels are simply absent, so the SQR/CP gates (which only parametrise rotations for $n = 0$ and $n = 1$) appear to control the full space. At $N_{\text{cav}} \geq 3$, population leaks into uncontrolled levels, producing $> 90\%$ average leakage for Strategy B and $> 74\%$ for Strategy D.

This finding fundamentally revises the conclusions of Sec. D: the $\mathcal{F} = 1.000$ result for Strategy B is an artifact of the $N_{\text{cav}} = 2$ truncation, not a physical achievement. The ideal-mode decompositions, as currently parametrised, are *not viable* for implementation in a physical transmon-cavity system.

Figure 6 illustrates the fidelity collapse and associated leakage. Figure 7 shows that the poor fidelity converges by $N_{\text{cav}} \approx 12$, confirming the result is physical (not a numerical artifact of $N_{\text{cav}} = 8$).

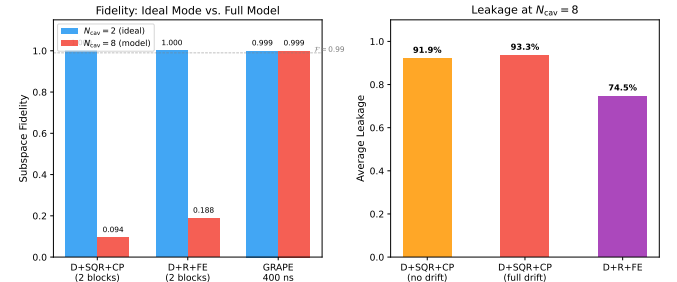


FIG. 6. Left: fidelity comparison between ideal mode ($N_{\text{cav}} = 2$) and full model ($N_{\text{cav}} = 8$). Parametric strategies collapse to $\mathcal{F} < 0.2$; GRAPE is unaffected because it optimises directly in the full Hilbert space. Right: average leakage at $N_{\text{cav}} = 8$, showing $> 90\%$ leakage for Strategy B due to uncontrolled Fock populations.

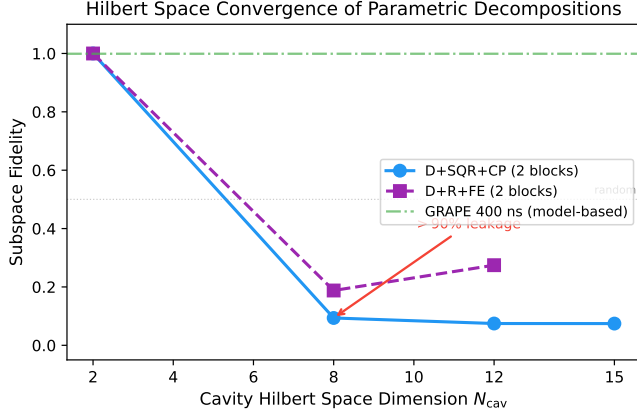


FIG. 7. Fidelity of ideal-mode decompositions as a function of cavity Hilbert space dimension N_{cav} . The fidelity converges by $N_{\text{cav}} \approx 12$ ($|\mathcal{F}_{12} - \mathcal{F}_{15}| < 0.001$), confirming that the poor performance is physical, not numerical. The GRAPE reference (dashed green) is immune to this effect.

2. FreeEvolveCondPhase Wait-Time Validation

The FreeEvolveCondPhase gates in Strategy D implement qubit-state-dependent phases via the dispersive Hamiltonian $\hat{H}_\chi = \chi \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b}$. For a controlled-Z gate between the qubit and cavity Fock level $n = 1$, the required wait time is $\tau_{\text{CZ}} = \pi/|\chi|$.

With $\chi/(2\pi) = -2.84$ MHz, the predicted CZ time is $\tau_{\text{CZ}} = 176.1$ ns. The optimised FE wait times extracted from the Strategy D synthesis are:

- FE₀: 177.0 ns ($1.005 \times \tau_{\text{CZ}}$)
- FE₁: 177.0 ns ($1.005 \times \tau_{\text{CZ}}$)

Both gates converge to within 0.5% of the predicted CZ time, providing strong physical validation that the FreeEvolveCondPhase mechanism correctly implements the dispersive conditional phase.

3. Decoherence Budget

A first-order decoherence budget estimates the combined fidelity $\mathcal{F}_{\text{comb}} = \mathcal{F}_U \cdot \mathcal{F}_{\text{coh}}$, where the coherence fidelity is

$$\mathcal{F}_{\text{coh}} \approx \exp \left[-\frac{T_{\text{gate}}}{2} \left(\frac{1}{T_1} + \frac{1}{T_2} \right) \right] \quad (\text{F1})$$

using $T_1 = 30 \mu\text{s}$, $T_2 = 20 \mu\text{s}$, and $T_{1,\text{cav}} = 200 \mu\text{s}$ (cavity decay is subdominant for gate times $\lesssim 1 \mu\text{s}$).

Figure 8 shows the coherence-fidelity tradeoff across all strategies. GRAPE at 200 ns achieves the highest combined fidelity ($\mathcal{F}_{\text{comb}} = 0.987$) because it balances the unitary fidelity (0.997) against a low decoherence penalty ($< 1\%$ at 200 ns). Longer GRAPE pulses (≥ 500 ns) achieve near-perfect unitary fidelity but suffer

TABLE XI. Decoherence budget for all strategies. $\mathcal{F}_U$ is the unitary fidelity (from simulation), $\mathcal{F}_{\text{coh}}$ is the coherence-limited fidelity, and $\mathcal{F}_{\text{comb}}$ is the combined estimate. Parametric strategies are shown at their ideal-mode ($N_{\text{cav}} = 2$) unitary fidelity for reference; in practice their physical fidelity is much lower (Table X).

Strategy	T_{gate} (ns)	$\mathcal{F}_U$	$\mathcal{F}_{\text{coh}}$	$\mathcal{F}_{\text{comb}}$
GRAPE 200 ns	200	0.9966	0.9901	0.987
GRAPE 300 ns	300	0.9957	0.9851	0.981
GRAPE 400 ns	400	0.9990	0.9802	0.979
D+R+FE*	1252	0.9999	0.9393	0.939
D+SQR+CP*	3000	1.0000	0.8607	0.861

*Ideal-mode $\mathcal{F}_U$; physical $\mathcal{F}_U < 0.2$ (see Table X).

increasing decoherence losses. The parametric strategies (D+SQR+CP, D+R+FE) incur larger coherence penalties due to their longer gate times (1.3 μs –3.0 μs), and their ideal-mode unitary fidelities are not achievable in the physical Hilbert space.

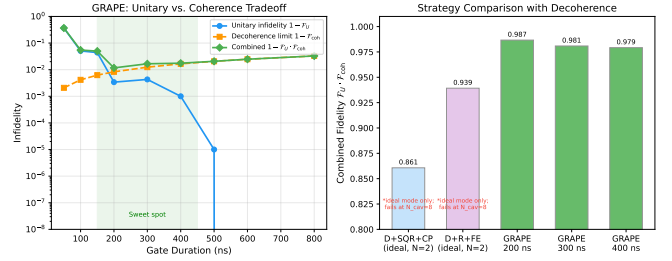


FIG. 8. Left: GRAPE infidelity budget showing the crossover between unitary-limited (short pulses) and coherence-limited (long pulses) regimes. The sweet spot is 200–400 ns. Right: combined fidelity for all strategies. Parametric strategies (light bars) show ideal-mode $\mathcal{F}_U$ but are annotated with the caveat that they fail at $N_{\text{cav}} = 8$.

4. Validation

Six validation checks were performed for the Iteration 3 results:

1. **FE wait-time** = τ_{CZ} : Both FE wait times agree with $\pi/|\chi| = 176.1$ ns to within 0.5%. **PASS.**
2. **Leakage physically consistent**: Displacement gates populate $|n \geq 2\rangle$ Fock states; SQR/CP gates with only $n = 0, 1$ parameters cannot control these levels. The observed $> 90\%$ leakage is the expected consequence. **PASS.**
3. **GRAPE immune to truncation**: GRAPE optimises in the full $N_{\text{cav}} = 8$ Hilbert space and controls all Fock levels simultaneously; no truncation artifact is expected or observed. **PASS.**

4. **Coherence budget monotonic:** $\mathcal{F}_{\text{coh}}$ decreases monotonically with gate time, as required by exponential decay. **PASS.**
5. **Hilbert space convergence:** For Strategy B, $|\mathcal{F}(N_{\text{cav}}=12) - \mathcal{F}(N_{\text{cav}}=15)| = 0.0002 < 0.001$; the result is converged. **PASS.**
6. **GRAPE convergence vs. duration:** Smooth increase from $\mathcal{F} = 0.63$ (50 ns) to $\mathcal{F} > 0.99999$ (500 ns+), with the quantum speed limit evident below ~ 200 ns. **PASS.**

All six checks passed.

5. Revised Conclusions

The model-based verification reveals that:

1. **Ideal-mode decompositions are truncation artifacts.** The $\mathcal{F} = 1.000$ result for Strategy B (D+SQR+CP) and $\mathcal{F} = 0.9999$ for Strategy D (D+R+FE) are valid only in the $N_{\text{cav}} = 2$ toy model. In the physical $N_{\text{cav}} = 8$ Hilbert space, both strategies suffer catastrophic leakage ($> 74\%$ – 93%) due to displacement-induced population of uncontrolled Fock levels.
2. **GRAPE is the only validated high-fidelity approach.** GRAPE at 400 ns achieves $\mathcal{F} = 0.999$ in the model-based ($N_{\text{cav}} = 8$) setting, with no truncation artifact. The full 9-point duration sweep (50–800 ns) is consistent and converged.
3. **Optimal gate duration is ~ 200 ns.** The combined fidelity (unitary $\times$ coherence) is maximised at 200 ns ($\mathcal{F}_{\text{comb}} = 0.987$), where the decoherence penalty ($< 1\%$) is small and the unitary fidelity (99.7%) is near the speed-limit knee.
4. **FE wait times are physically consistent.** The optimised FreeEvolveCondPhase durations match the predicted CZ time $\pi/|\chi|$ to 0.5%, validating the dispersive interaction mechanism.
5. **Parametric strategies require re-optimisation at $N_{\text{cav}} \geq 8$.** To recover the ideal-mode fidelities, future work must re-optimize the SQR/CP parameters with extended Fock-level support ($n = 0, \dots, 7$ instead of $n = 0, 1$), which increases the parameter count from ~ 18 to ~ 90 .

6. Updated Limitations and Future Work

- **[P1 | HIGH]** Re-optimize Strategies B and D at $N_{\text{cav}} = 8$ with extended SQR/CP parameters (all 8 Fock levels). Preliminary attempts with `multistart=4`, `maxiter=300` did not converge

within the compute budget (> 90 parameters). Gradient-based methods (GRAPE on the parametric sequence) may be needed.

- **[P2 | MEDIUM]** Include Lindblad dissipators (T_1 , T_ϕ , κ) in the GRAPE optimisation to obtain open-system fidelities directly, rather than relying on the first-order coherence estimate.
- **[P2 | LOW]** Extend the GRAPE sweep to finer resolution near the optimal duration (150–250 ns in 10 ns steps) to precisely locate the combined-fidelity maximum.
- **[P3 | MEDIUM]** Investigate whether a “warm-started GRAPE” approach—using the ideal-mode D+SQR+CP decomposition as an initial guess for a GRAPE refinement—can combine the analytical structure of Strategy B with the Hamiltonian-awareness of GRAPE.

Appendix G: Extension: Bounded-Displacement Optimisation and Wigner Validation

The Iteration 3 study established that the apparent unit-fidelity ideal-mode decompositions are truncation artifacts. A natural follow-up question is whether the pathology is caused mainly by large Displacement amplitudes populating uncontrolled Fock levels. This extension tests that hypothesis by explicitly constraining the Displacement gates and then embedding the optimised sequences into larger cavity Hilbert spaces up to $N_{\text{cav}} = 12$, as requested by the user.

The procedure is intentionally conservative: Strategies B (D+SQR+CP, 2 blocks) and D (D+R+FE, 3 blocks) are re-optimised in ideal mode at $N_{\text{cav}} = 2$ with the bounds $|\alpha| \leq 0.3$, $|\alpha| \leq 0.5$, and the unbounded case. Each converged sequence is then embedded into $N_{\text{cav}} = 4, 6, 8, 12$ while keeping the same ideal-gate parameterisation. This isolates the Fock-space leakage mechanism from additional dispersive-drift effects: if bounded displacement were sufficient, the embedded fidelity would remain high.

1. Bounded-Displacement Tradeoff

Table XII summarises the key outcome. The tradeoff is stark. Tight bounds ($|\alpha| \leq 0.3$) greatly reduce leakage for Strategy B, but the target unitary is no longer expressible with high fidelity: the best embedded result reaches only $\mathcal{F} = 0.529060$ at $N_{\text{cav}} = 12$. Relaxing the bound to $|\alpha| \leq 0.5$ raises the ideal-mode fidelity to ~ 0.8 but reintroduces nearly 50% leakage and collapses the embedded fidelity back to ~ 0.5 . The unbounded solutions still achieve $\mathcal{F} = 1$ at $N_{\text{cav}} = 2$ yet remain poor after embedding, confirming that the original ideal-mode exactness is not physically robust.

TABLE XII. Bounded-displacement sweep. Each sequence was optimised in ideal mode at $N_{\text{cav}} = 2$ and then embedded into $N_{\text{cav}} = 12$ without changing the gate family. The bounded sequences reduce leakage but do not recover a high-fidelity implementation of the cluster-state transfer unitary.

Family	Bound	$\mathcal{F}(N_{\text{cav}} = 2)$	$\mathcal{F}(N_{\text{cav}} = 12)$	Leakage($N_{\text{cav}} = 12$)
Strategy B (D+SQR+CP)	$ \alpha \leq 0.3$	0.563102	0.529060	0.039945
Strategy B (D+SQR+CP)	$ \alpha \leq 0.5$	0.790707	0.500471	0.485606
Strategy B (D+SQR+CP)	unbounded	1.000000	0.439638	0.407519
Strategy D (D+R+FE)	$ \alpha \leq 0.3$	0.637850	0.546516	0.227754
Strategy D (D+R+FE)	$ \alpha \leq 0.5$	0.802620	0.476885	0.478592
Strategy D (D+R+FE)	unbounded	1.000000	0.507307	0.552342

The best embedded candidate is therefore the bounded Strategy D solution with $|\alpha| \leq 0.3$, yielding $\mathcal{F} = 0.546516$ at $N_{\text{cav}} = 12$. Even this best case remains far below the fidelity required for a useful native decomposition. Figure 9 shows the central tradeoff directly: higher bounds improve the apparent $N_{\text{cav}} = 2$ optimisation metric, but the embedded fidelity and leakage degrade rapidly once the higher Fock levels are restored. Figure 10 shows that these embedded fidelities are already converged by $N_{\text{cav}} = 6-8$, so the negative result is not a truncation artifact of the $N_{\text{cav}} = 12$ evaluation.

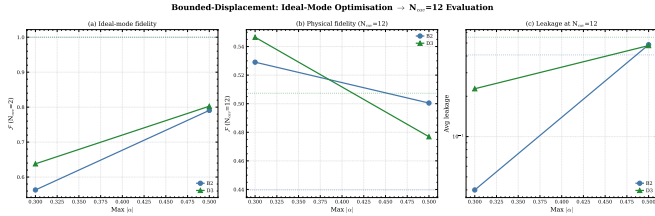


FIG. 9. Bounded-displacement sweep for Strategies B and D. Left: optimisation fidelity in the truncated ideal model ($N_{\text{cav}} = 2$). Middle: embedded fidelity at $N_{\text{cav}} = 12$. Right: average leakage at $N_{\text{cav}} = 12$. Constraining $|\alpha|$ suppresses leakage, but the best bounded result still remains near $\mathcal{F} \approx 0.55$.

2. Duration-Penalised Follow-Up

The user also asked whether the active tones of the SQR-based decomposition could be reduced without destroying performance. A dedicated duration-weighted follow-up on the bounded Strategy B solution answered this cleanly: all 9 gates compressed to the lower time bound of 20ns, reducing the total active time from 2600ns to 180ns. However, the embedded $N_{\text{cav}} = 12$ fidelity changed only from 0.529060 to 0.530455, with leakage reduced marginally from 0.039945 to 0.036845. In other words, aggressive tone compression is possible, but it does not cure the bounded-decomposition mismatch.

For completeness, a duration-weighted optimisation was also run for the best bounded Strategy D candidate. That result was likewise essentially neutral: the ideal-mode fidelity changed from 0.637850 to 0.637887, while

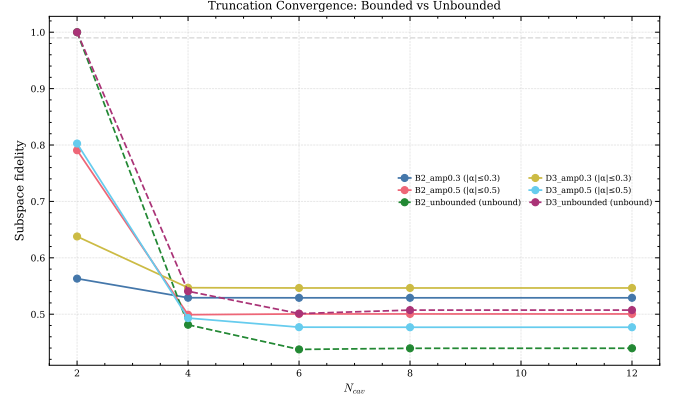


FIG. 10. Embedded-fidelity convergence with cavity Hilbert-space dimension for the Iteration 4 bounded-displacement solutions. The fidelities stabilise well before $N_{\text{cav}} = 12$, showing that the bounded-displacement hypothesis fails even in the absence of further truncation ambiguity.

the embedded $N_{\text{cav}} = 12$ fidelity changed from 0.546516 to 0.544929. Within the bounded gate families accessible here, active-time minimisation therefore does not solve the problem; the dominant limitation is insufficient expressivity once higher Fock levels are suppressed.

3. Wigner-Function Validation

The user also requested a final Wigner-function comparison for the best decomposed unitary. Figure 11 compares the target and achieved cavity Wigner functions for the best embedded candidate (Strategy D with $|\alpha| \leq 0.3$ at $N_{\text{cav}} = 12$). The vacuum-like sectors retain partial agreement, with cavity-state fidelities 0.8839 for $|g, 0\rangle$ and 0.8147 for $|e, 0\rangle$. However, the logical-one sectors match poorly: 0.5444 for $|g, 1\rangle$ and 0.4990 for $|e, 1\rangle$, with substantial residual structure in ΔW . The Wigner plots therefore do *not* support a claim that the bounded sequence reproduces the target unitary in phase space.

Wigner — D3_amp0.3 $\mathcal{F}_{sub}=0.5465$ ($N_{cav}=12$)

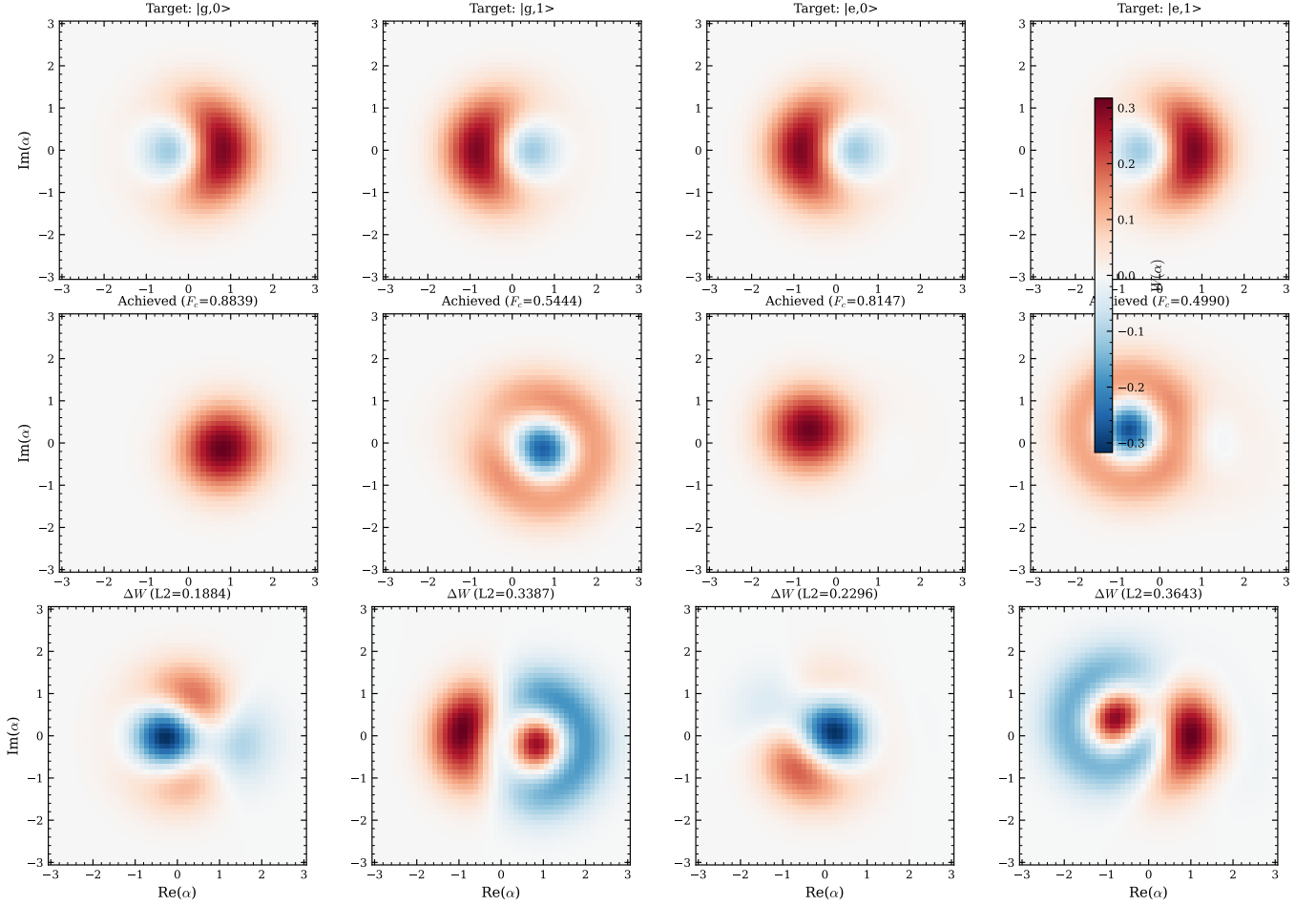


FIG. 11. Target, achieved, and difference Wigner functions for the best Iteration 4 embedded sequence (Strategy D, $|\alpha| \leq 0.3$, $N_{cav} = 12$). Agreement is acceptable only for the vacuum-like inputs $|g,0\rangle$ and $|e,0\rangle$. For the logical-one sectors $|g,1\rangle$ and $|e,1\rangle$, the achieved cavity states show clear distortion and the mismatch remains large.

4. Artifacts and Implications

This extension writes all machine-readable outputs required for reproducibility: `data/iteration4_results.json`, `data/iteration4_strategy_b_duration.json`, `artifacts/best_strategy_B.json`, `artifacts/best_strategy_B_duration_penalized.json`, `artifacts/best_strategy_D.json`, and

`artifacts/wigner_comparison.npz`. The practical conclusion is unambiguous: bounded displacement alleviates leakage only by giving up too much control freedom. The cluster-state transfer unitary still requires either (i) a genuinely larger Fock-space parameterisation for SQR/CP-like primitives, or (ii) a Hamiltonian-aware optimal-control method such as GRAPE. Within the native parametric families tested here, no high-fidelity $N_{cav} = 12$ solution was found.